

Davide Capone

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SUMMARY

Machine Learning Engineer specializing in computer vision and AI systems for industrial applications. Experienced in developing end-to-end machine learning pipelines, including object detection, multi-object tracking, and multimodal document retrieval systems. Focused on building reliable AI solutions for real-world environments.

EXPERIENCE

Data Scientist

Danieli S.p.A.

Jul 2025 – Present

Buttrio (UD), Italy

- Designed and developed AI-driven solutions for industrial and business applications, integrating machine learning, computer vision, and data analytics.
- Implemented multimodal document intelligence systems using Retrieval-Augmented Generation (RAG) and Vision-Language Models (VLMs) for semantic search across technical documentation.
- Developed computer vision pipelines for object detection to support automation and monitoring in industrial environments.
- Built business intelligence dashboards and analytics workflows to support operational insights and data-driven decision-making.

Machine Learning Engineer (Intern)

Danieli Automation S.p.A.

Dec 2024 – May 2025

Buttrio (UD), Italy

- Designed and developed an end-to-end multi-object tracking system for industrial steel production environments, addressing real-world operational constraints.
- Evaluated multiple state-of-the-art deep learning object detection models to identify optimal trade-offs between accuracy, robustness, and inference performance.
- Integrated selected detection models with Kalman filtering and data association techniques to implement a robust multi-object tracking pipeline.
- Optimized and deployed trained models in ONNX format to enable efficient and scalable industrial inference.

Machine Learning Engineer (Intern)

Danieli Automation S.p.A.

Oct 2022 – Dec 2022

Buttrio (UD), Italy

- Applied advanced computer vision techniques, including camera calibration and homography estimation, to enable accurate object detection and spatial localization in industrial environments.
- Designed, trained, and evaluated convolutional neural network (CNN) models for object detection and localization under real-world industrial constraints.

EDUCATION

University of Trieste

Master of Science in Data Science and Scientific Computing

Trieste, Italy

Aug 2022 – May 2025

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Optimization

University of Udine

Bachelor of Science in Internet of Things, Big Data, and Machine Learning

Udine, Italy

Aug 2019 – May 2022

- Relevant coursework: Algorithms and Data Structures, Machine Learning, Statistics, Cloud Computing

PROJECTS

GAPC – Asteroid Observation Catalog

Web Application Project

Python, Django, Databases

- Developed a Django application to catalog and manage asteroid observations following astronomical data standards.
- Designed database schemas and backend logic to efficiently store, retrieve, and analyze scientific metadata.
- Implemented data management features to support structured analysis and research workflows.

YOH024 – Sound Event Detection Model

Deep Learning Project

Python, PyTorch, Audio Processing

- Designed and implemented a deep learning model for sound event detection inspired by the YOLO (You-Only-Hear-Once) paradigm.
- Extended the original YOHO architecture by integrating improvements from later YOLO versions to enhance efficiency and detection accuracy.
- Evaluated model performance on standard audio segmentation datasets and benchmarked results against the original YOHO implementation.

Weighted Matrix Factorization Recommender

Machine Learning Project

Python, Recommender Systems

- Developed a movie recommendation system based on weighted matrix factorization techniques.
- Implemented and compared optimization strategies including Stochastic Gradient Descent (SGD) and Weighted Alternating Least Squares (WALS).
- Analyzed model performance to improve recommendation accuracy and system efficiency.

Traffic Lights Optimization

Reinforcement Learning Project

Python, Reinforcement Learning, Simulation

- Applied reinforcement learning methods, including Value Iteration and Policy Iteration, to optimize traffic light control policies.
- Modeled traffic environments and simulated agent–environment interactions to reduce queue lengths and waiting times.
- Evaluated learned policies through simulation-based performance analysis.

TECHNICAL SKILLS

Programming: Python, C++, C, SQL, R, MATLAB

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, Object Detection, Multi-Object Tracking, Retrieval-Augmented Generation (RAG), Vision-Language Models (VLMs)

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Polars, Django

MLOps & Deployment: Model Evaluation, Hyperparameter Tuning, ONNX Export, Inference Optimization, Docker

Data & Analytics: Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA), Business Intelligence, Data Visualization

Tools: Git, SQL Databases, Power BI

CERTIFICATIONS

- Introduction to Deep Learning & Neural Networks with Keras — IBM (TensorFlow, Keras, Neural Networks)
- Machine Learning in Production — DeepLearning.AI (MLOps, Deployment, Monitoring)
- Machine Learning with Python — IBM (Python, Scikit-learn, Feature Engineering)